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Wireless:

“Wi-Fi Is The ONLY TECHNOLOGY That Can Prevail THE LONGEST”

With the Internet of Things (IoT) being the buzz word these days, umpteen choices of ultra-modern wireless technologies have cropped up. How do these stand up to Wi-Fi? Venkat Mattela, founder, chairman and chief executive officer, Redpine Signals Inc., speaks with Dilin Anand from EFM

Q. What is the most robust wireless technology available to engineers?

A. We started Redpine Signals in 2012 with an intent to converge nine wireless technologies on the same silicon die. Fast forward 15 years, a couple of wireless protocols like DVB-H and UWB did not make it even. Ethernet is the best proven and known networking technology but the equivalent to it in wireless is wireless local area network (WLAN), or Wi-Fi. It is the only wireless technology developed for networking. Wi-Fi has been able to deliver good throughput at much lower power consumption than any other wireless technology. It provides the required throughput and power profile at the right price point for varied applications in the IoT space.

Q. Do you think it will survive over the long term?

A. Wi-Fi is the only technology that can prevail the longest because it is the easiest and cheapest way to implement wireless technology in a device. Wi-Fi is the only tech that can give you a US\$1 wireless chip and below 100-micro-ampere level of standby associated state.

This was not possible five years ago with Wi-Fi, which is why other technologies came into the picture. For vehicular communication, 802.11p is there, but ZigBee will thrive in the long term. Technologies like LoRa may not be viable in the long term because of throughput-related challenges.

Q. How has Wi-Fi evolved for autonomous cars?

A. There are two aspects: safety-critical communication and entertainment-purpose communication. For the former like collision avoidance, you need robust communication. Also, the connection between the wireless onboard unit and the roadside unit must be very deterministic and have low latency. This is why Dedicated Short-Range Communications (DSRC) 802.11p came into the picture. It is a modification of the existing Wi-Fi technology with a software stack and hardware changes that only guarantees determinism as compared to regular Wi-Fi.

Q. Could you elaborate a bit about 802.11p?

A. 802.11p is basically a V2V or V2X technology. This wireless technology is used to connect a car to roadside units, which connect to the infrastructure. It functions almost like Wi-Fi connecting to an access point, except that OBU Wi-Fi moves much faster than regular Wi-Fi station cards. This requires extra signal processing to provide proper reception for robust communication.

Q. What do engineers need to know about Wi-Fi?

A. The main challenge with Wi-Fi is latency. Wireless also has an intrinsic problem that you can block it with external interference.

On the other hand, the level of validation of wireless technologies today ensures that you can reliably use applications that consume a lot of data like a video call from a village or from below the ground. The level of maturity and mitigation knowledge that have gone into this technology is noteworthy. This makes wireless technology a default part of any new device.

Q. Vendors keep coming out with new low-power solutions. Where will this end, or will it?

A. The concept of low power is psychological. Ten years ago, 100 milli-watts was considered low power. Today, one milli-watt is low power. The aim for low power should be the ability to run without power. It should be able to harvest what little energy it needs to run so that an external power source is not required. This is what we have achieved with our Wi-Fi circuits for certain applications.

Q. How were you able to use an unsuspecting technology for a completely-unrelated application?

A. Surprisingly, Wi-Fi itself has turned out to be a great example of unexpected technology that can be used for energy harvesting. It was designed for networking with plugged-in devices. However, we were able to do an energy-harvesting project using it by choosing an application that did not require a heavy dose of data in frequent intervals. **EFM**